Biology 336 Formula Sheet – Final Exam Version

Haploid selection:

$$p[t+1] = \frac{W_A\ p[t]}{W_A p[t] + W_a q[t]}$$

$$\Delta p[t] = p[t+1] - p[t] = \frac{(W_A - W_a)\ p[t]\ q[t]}{W_A\ p[t] + W_a\ q[t]}$$

Allele frequency after several generations:

$$\frac{p[t]}{q[t]} = \frac{W_A^t\ p[0]}{W_a^t\ q[0]} \quad \text{or equivalently} \quad p[t] = \frac{W_A^t\ p[0]}{W_A^t\ p[0] + W_a^t\ q[0]}$$

Mean fitness:

$$\bar{W} = W_A p[t] + W_a q[t]$$

Diploid selection:

Genotype frequencies when in "Hardy-Weinberg proportions":

$$x_{AA}[t] = p[t]^2,\ \ x_{Aa}[t] = 2p[t]q[t],\ \ x_{aa}[t] = q[t]^2$$

Genotype frequencies after selection

$$x_{AA}^s = \frac{W_{AA} x_{AA}[t]}{\bar{W}},\ \ x_{Aa}^s = \frac{W_{Aa} x_{Aa}[t]}{\bar{W}},\ \ x_{aa}^s = \frac{W_{aa} x_{aa}[t]}{\bar{W}}$$

where the mean fitness is:

$$\bar{W} = W_{AA} x_{AA}[t] + W_{Aa} x_{Aa}[t] + W_{aa} x_{aa}[t]$$

Allele frequency in the next generation

$$p[t+1] = \frac{W_{AA}\ p[t]^2 + W_{Aa} p[t] q[t]}{W_{AA}\ p[t]^2 + W_{Aa} 2\ p[t] q[t] + W_{aa}\ q[t]^2}$$

Formula for polymorphic equilibrium (used only for Heterozygote advantage or disadvantage): $\hat{p} = \frac{W_{Aa} - W_{aa}}{2W_{Aa} - W_{AA} - W_{aa}}$

Mutations:

Mutation alone (no selection, haploid or diploid model): $p[t+1] = (1-\mu)\ p[t] + \nu\ (1 - p[t])$

Equilibrium: $\hat{p} = \frac{\nu}{\mu + \nu}$

Haploid model: with mutations and selection ($W_A = 1,\ \ W_a = 1 - s$)*, the deleterious allele "a" approaches:* $\hat{q} \approx \frac{\mu}{s}$

Diploid model: with mutations and selection, the deleterious allele "a" approaches:

$\hat{q} \approx \frac{\mu}{hs}$ *if h>0* *and* $\hat{q} \approx \sqrt{\frac{\mu}{s}}$ *if h=0 (i.e., if "a" is recessive to "A")*

At the diploid mutation-selection balance equilibrium ($\hat{q} \approx \frac{\mu}{hs}$), the mean fitness is $\bar{W} = 1 - 2\mu$, which does not depend on the strength of selection.

Genetic Drift:

- The chance that a neutral allele becomes fixed is its initial frequency *p*.
- The time for a neutral allele initially in a single copy to become fixed: ~2*N* generations in a haploid population; 4*N* generations in a diploid population.

Expected heterozygosity, $H[t]$, the chance that two randomly chosen alleles differ, is a measure of genetic variation.

$$H[0] = 2pq$$

$$H[t] = \left(1 - \frac{1}{2N}\right) H[t-1]$$

Expected homozygosity, $F[t]$, is the chance that two randomly chosen alleles are the same, $F[t] = 1 - H[t]$.

Haldane's fixation probability: 2*s* in haploids and *2hs* in a diploid (assuming *h*>0).

Two-locus model:

Disequilibrium, D (also known as linkage disequilibrium or gametic-phase disequilibrium), measures genetic associations among two loci.

$$D[t] = x_{AB}[t]\, x_{ab}[t] - x_{Ab}[t]\, x_{aB}[t]$$

Recombination acts to break down linkage disequilibrium. Meiosis with recombination changes the chromosome frequencies by:

$$x_{AB}^{M} = x_{AB} - r\,D \qquad x_{Ab}^{M} = x_{Ab} + r\,D$$
$$x_{aB}^{M} = x_{aB} + r\,D \qquad x_{ab}^{M} = x_{ab} - r\,D$$

When selection is absent (all $W_{ij} = 1$), genetic associations (D) decay by a factor $(1 - r)$ each generation:

$$D[t] = (1-r)D[t-1]$$

This means that over a number of generations, t, genetic associations decay according to:

$$D[t] = (1-r)^t\, D[0]$$

Quantitative Genetics:

Phenotypic variance, $V_P = V_G + V_E$, with effects due to genetics (G) and environment.

Genetic variance, $V_G = V_A + V_D + V_I$, including additive (A), dominance (D) and epistatic (or interaction, I) effects of alleles.

Heritability (=narrow sense heritability): $h^2 = \frac{V_A}{V_P}$

The breeder's equation: $R = h^2 S$

where S is the selection differential and R is the predicted response to selection in the next generation.

Spatial model:

Differentiation of two subpopulations is measured by $F_{ST} = \sigma_S^2/\sigma_T^2$, the variance in allele frequency among sub-populations (σ_S^2) relative to the total variance in allele frequency if population structure were ignored (σ_T^2). Also equal to $F_{ST} = (H_T - \bar{H}_S)/H_T$ where

- *H_T = <u>expected</u> heterozygosity when drawing alleles randomly from the entire population.*
- *H_S = <u>expected</u> heterozygosity in each subpopulation with its unique allele frequencies ($\bar{H}_S$ is the average of these).*